

5-state EKF: a' as a measured state (y_2 update, no bolt-on)

DRAFT — route follows `5state_aprime_var_esti.tex`; Sec. 2 replaces the β -EWMA of a'_{hdm} with a Kalman update of a' .

0. Definitions

$$h[k] = h_d[k] - \delta h[k], \quad \Delta h_d[k] = h_d[k+1] - h_d[k], \quad \Delta H_d[k] = h_d[k] - h_d[k-d]$$

$$a_d[k] := a_h(h_d[k]), \quad a'[k] := \left. \frac{da_h}{dh} \right|_{h_d[k]}, \quad a_r[k] := a_h(h[k]) - a_d[k]$$

$$\delta h[k+1] = \lambda_c \delta h[k] - F_{dh}[k] e_{ad}[k] - \varepsilon_h[k]$$

a_d = gain at the desired height (deterministic, trajectory-driven); a' = its slope; a_r = residual gain from the tracking error; ΔH_d = known d -step commanded sweep. $\delta h_1 := \delta h[k-d]$ (oldest, measured), $\delta h_2 := \delta h[k-1]$, $\delta h_3 := \delta h[k]$.

1. True

First-order Taylor of the gain at the actual (perturbed) height:

$$a_h(h[k]) = a_h(h_d[k] - \delta h[k]) \approx a_d[k] - a'[k] \delta h[k]$$

$$a_r[k] = -a'[k] \delta h[k], \quad a'[k+1] \approx a'[k]$$

The desired-height gain drifts with the commanded sweep:

$$a_d[k+1] = a_d[k] + a'[k] \Delta h_d[k]$$

The second measurement reads the *actual* gain d steps ago. Substituting $a_d[k-d] = a_d[k] - a' \Delta H_d[k]$ (from the sweep, a' slow) and $\delta h[k-d] = \delta h_1$:

$$y_2[k] = a_h(h[k-d]) = a_d[k-d] - a' \delta h[k-d] + \nu_2[k]$$

$$y_2[k] = a_d[k] - a'[k] (\Delta H_d[k] + \delta h_1[k]) + \nu_2[k]$$

a' enters y_2 *explicitly*, multiplying the total height offset $\Delta H_d + \delta h_1$ (commanded sweep + thermal tracking error). This is the relation the var-channel used implicitly; here it is kept as a measurement.

2. Estimator (a' inside the KF)

State (5×1):

$$x = [\delta h_1, \delta h_2, \delta h_3, a_d, a']^\top$$

Predict (deterministic map; $\Delta h_d, \Delta H_d$ known from the trajectory):

$$\begin{aligned} \delta h_1[k+1] &= \delta h_2[k], & \delta h_2[k+1] &= \delta h_3[k], & \delta h_3[k+1] &= \lambda_c \delta h_3[k], \\ a_d[k+1] &= a_d[k] + \hat{a}'[k] \Delta h_d[k], & a'[k+1] &= a'[k]. \end{aligned}$$

Measurement ($y_1 =$ delayed position error, $y_2 =$ gain gauge):

$$y_1[k] = \delta h_1[k] + n_x[k], \quad y_2[k] = a_d[k] - a'[k](\Delta H_d[k] + \delta h_1[k]) + \nu_2[k]$$

$$H_1 = [1, 0, 0, 0, 0]$$

$$H_2[k] = [-\hat{a}'[k], 0, 0, 1, -(\Delta H_d[k] + \delta \hat{h}_1[k])]$$

Update (standard EKF; $H = [H_1; H_2]$, $R = \text{diag}(\sigma_n^2, R_2)$):

$$K[k] = P^- H^\top (H P^- H^\top + R)^{-1}, \quad x^+ = x^- + K(y - Hx^-)$$

Why a' is now genuinely in the KF. The gain-row of K is

$$K_5 = \frac{[P^- H^\top]_5}{S} = \frac{P_{55}^- H_2(5) + P_{54}^- H_2(4) + \dots}{S}, \quad H_2(5) = -(\Delta H_d + \delta \hat{h}_1) \neq 0.$$

a' carries its *own* measurement entry $P_{55}^- H_2(5)$ (direct correction), not only the cross-covariance P_{54}^- . This replaces the esti recursion $a'[k+1] = \beta a'[k] + (1-\beta) a'_{hdm}[k]$ (a β -EWMA of a variance ratio computed *outside* the filter): a' now updates from the y_2 innovation like any other state.

3. Error dynamics

$e_h = \delta h - \delta \hat{h}$, $e_{a_d} = a_d - \hat{a}_d$, $e_{a'} = a' - \hat{a}'$. Predict (before correction):

$$e_{a_d}[k+1] = e_{a_d}[k] + \Delta h_d[k] e_{a'}[k], \quad e_{a'}[k+1] = e_{a'}[k],$$

so an a' error feeds the level error at rate Δh_d (the sweep coupling). Linearizing the y_2 innovation about the current estimates:

$$\nu_2 = e_{a_d} - (\Delta H_d + \delta \hat{h}_1) e_{a'} - \hat{a}' e_{h,1} + \nu_2^{\text{noise}}.$$

The closed loop is the usual EKF contraction

$$e[k+1] = (I - K[k]H[k]) F_e[k] e[k] + q[k]$$

with $e = [\delta h_1, \delta h_2, \delta h_3, e_{a_d}, e_{a'}]^\top$. Because $H_2(5) \neq 0$, the correction $(I - KH)$ acts on $e_{a'}$ *directly* through the y_2 innovation — the mechanism the bolt-on lacked.

4. F_e and H

$$F_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dh}[k] & 0 \\ 0 & 0 & 0 & 1 & \Delta h_d[k] \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad H[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ -\hat{a}'[k] & 0 & 0 & 1 & -(\Delta H_d[k] + \delta \hat{h}_1[k]) \end{bmatrix}$$

Time-varying entries: $F_e(3, 4) = -F_{dh}[k]$, $F_e(4, 5) = \Delta h_d[k]$ (sweep coupling, predict), and the whole of $H_2[k]$ (measurement coupling). Row 5 of F_e is a pure integrator; a' acquires dynamics only through the measurement, not a process pole.

5. Observability and caveats

Two channels in one entry. $H_2(5) = -(\Delta H_d + \delta \hat{h}_1)$: the commanded sweep ΔH_d (deterministic, present only in motion) and the thermal tracking error $\delta \hat{h}_1$ (stochastic, present in holds). Both routes to a' live in the single measurement entry.

Bilinear EKF. y_2 contains the product $a' \delta h_1$; both $H_2(1) = -\hat{a}'$ and $H_2(5) = -(\Delta H_d + \delta \hat{h}_1)$ come from it, so a' and δh_1 are coupled in the same measurement. Identifiability (does an innovation load onto $e_{a'}$ or $e_{h,1}$?) is set by the covariance and must be checked — in a pure hold ($\Delta H_d = 0$) both are read from the same thermal product.

Hold floor unchanged. $\Delta H_d = 0 \Rightarrow H_2(5) = -\delta \hat{h}_1$, so a' observability scales with the thermal amplitude $\sim \sqrt{\text{Var}(\delta h)}$ over a finite dwell — a CRLB floor, independent of this reformulation.

Not self-referential. $y_2 = a_{xm}$ is a real measurement carrying $-a'_{\text{true}} \delta h$; the innovation delivers genuine truth to a' , unlike the var-channel's self-constructed $\hat{a}_r$ (which returns $\hat{a}'$).

6. Open (to fill next)

Closed-loop variance (Lyapunov on e), Q_{55} process noise, and the R_2 / y_2 -noise model are deferred. The identifiability check (Sec. 5) and a hold/osc simulation come before locking these.